25900518 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMBG201 - MICROBIAL GENETICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minitues **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	Which strain of Streptococcus pneumoniae caused the death of mice in Griffith's experiment?	[An] / [CO1, CO2, CO3]
2.	Give the limitations of Griffith's experiment.	[K] / [CO1, CO2, CO3]
3.	What are intercalating agents? Give one example.	[K] / [CO3]
4.	State one advantage of rolling circle replication.	[K] / [CO1, CO2, CO3]
5.	Mention the steps involved in generalised transduction.	[K] / [CO2]
6.	Define Okazaki fragments.	[K] / [CO1, CO2, CO3]
7.	What is genetic recombination.	[K] / [CO2]
8.	What are transposons?	[K] / [CO2]

9.	Define excision repair in bacteria.	[K] / [CO3]
10.	What is the key difference between generalised and specialised transduction.	[K] / [CO2]
11.	What are the steps of gene expression in bacteria.	[K] / [CO2]
12.	What is the purpose of the Ames test?	[K] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13.	What are lethal mutations? How can conditional mutants help in studying lethal genes?	[K] / [CO3]
14.	Explain the mechanism of F^+–F^- conjugation in bacteria. What is the role of the F plasmid in this process?	[U] / [CO1, CO2, CO3]
15.	Explain the mechanism by which gamma rays cause DNA mutations.	[U] / [CO3]
16.	Explain composite transposons with example.	[K] / [CO1, CO2, CO3]
17.	Illustrate the sequence of events that occur when E. coli is transferred from a glucose-rich medium to a lactose-only medium.	[U] / [CO2]
18.	Distinguish between substitution, insertion, and deletion mutations with one example each.	[U] / [CO3]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

19.	What are the applications of plasmids?	[K] / [CO1, CO2, CO3]
20.	Explain the key components involved in the induction and repression of gene expression in prokaryotes.	[U] / [CO2]

[10x1 = 10]